

MedTrack_DV Dashboard Storyboard

Module 4: Dashboard Planning and Prototyping

This storyboard defines the planned Tableau workbook structure, dashboard layouts, filters, navigation, and actions for the Hospital Operations & Patient Analytics Dashboard.

Dashboards Planned

- Hospital Overview
- Patient Flow
- Department Analytics
- Resource Utilization

Global Controls

Date Range, Hospital, Department, Region, Admission Type, Discharge Status, and Reset Filters.

Primary Data Source

data/processed/hospital_final_dataset.xlsx - Final Dataset sheet

Dashboard 1 - Hospital Overview

Executive view for admissions, occupancy, readmissions, and overall hospital performance.

MedTrack DV	Date Range Hospital Department Region			
Navigation Overview Patient Flow Department Resources	Total Admissions KPI tile	Occupancy Rate KPI tile	Avg Length of Stay KPI tile	Readmission Rate KPI tile
	Admissions Trend Monthly line chart	Occupancy Trend Monthly line chart	Readmission Trend Monthly line chart	Admissions by Type Donut chart
	Admissions by Department Bar chart	Admissions by Region Map or filled region	Admissions vs Discharges Combo bar chart	Top Hospitals Ranked bar chart

Planning Notes

- Primary users: hospital administrators and healthcare managers.
- Default view should show full year 2024 and all hospitals.
- KPI tiles should use teal, green, blue, and magenta accents for quick scanning.

Dashboard 2 - Patient Flow

Operational view for admission trends, discharge tracking, movement, average stay, and peak patient load.

MedTrack DV

Date Range | Hospital | Department | Region

Navigation

Overview

Patient Flow

Department

Resources

Admissions KPI tile	Discharges KPI tile	Avg Stay KPI tile	Peak Month KPI tile
Daily/Monthly Admissions Line chart	Discharges by Status Stacked bar	Admission Type Mix Donut chart	Length of Stay Distribution Histogram
Patient Movement Admission to discharge timeline	Peak Load by Month Heatmap	Flow by Department Sankey-style bar	Age Group Flow Stacked bar

Planning Notes

- Use admission_date and discharge_date for flow timing.
- Clicking a month should filter length-of-stay and department movement visuals.
- Highlight emergency admissions separately because they affect capacity planning.

Dashboard 3 - Department Analytics

Department comparison view for volume, readmissions, efficiency, and treatment capacity.

MedTrack DV	Date Range Hospital Department Region			
Navigation Overview Patient Flow Department Resources	Departments Count tile	Best Efficiency KPI tile	Highest Readmission KPI tile	Highest Volume KPI tile
	Volume by Department Horizontal bar	Readmission by Department Bar chart	Efficiency Score Ranked bar	Avg Stay by Department Bar chart
	Capacity by Department Grouped bar	Hospital x Department Heatmap	Discharge Outcomes Stacked bar	Department Detail Text/KPI panel

Planning Notes

- Sort departments by total admissions by default.
- Use Department Efficiency Score as the main comparison metric.
- Selecting a department should filter hospital and resource details.

Dashboard 4 - Resource Utilization

Resource planning view for beds, staff, equipment, capacity, and availability.

MedTrack DV	Date Range Hospital Department Region			
Navigation Overview Patient Flow Department Resources	Bed Utilization KPI tile	Equipment Utilization KPI tile	Staff Coverage KPI tile	Available Capacity KPI tile
	Bed Utilization by Dept Bar chart	Staff Available vs Required Grouped bar	Equipment Use Bar chart	Capacity Trend Monthly line
	Hospital Capacity Ranked bar	Resource Availability Heatmap	Utilization Scatter Beds vs staff	Planning Notes Text panel

Planning Notes

- Use red/magenta warnings when utilization or staffing becomes risky.
- Hospital and department filters should drive the resource views.
- This dashboard supports capacity planning decisions.

Filters, Navigation, and Dashboard Actions

Interaction	Source	Target	Purpose
Dashboard navigation	Sidebar buttons	All dashboards	Move between four views
Global filters	Date, hospital, department, region	All dashboards	Consistent slicing
Department selection	Department charts	Overview and resources	Compare selected department
Month selection	Trend charts	Patient flow	Investigate peak patient load
Hospital selection	Hospital charts	Department and resource views	Hospital-level drilldown

Prototype Acceptance Checklist

- Each dashboard uses the Final Dataset sheet from hospital_final_dataset.xlsx.
- Global filters are visible and apply across all four dashboards.
- Navigation buttons move between Hospital Overview, Patient Flow, Department Analytics, and Resource Utilization.
- KPI tiles match values from the Overall KPIs sheet.
- Department and hospital drilldowns respond to selected marks.
- Workbook is saved from Tableau as dashboard/medtrack_prototype.twbx.